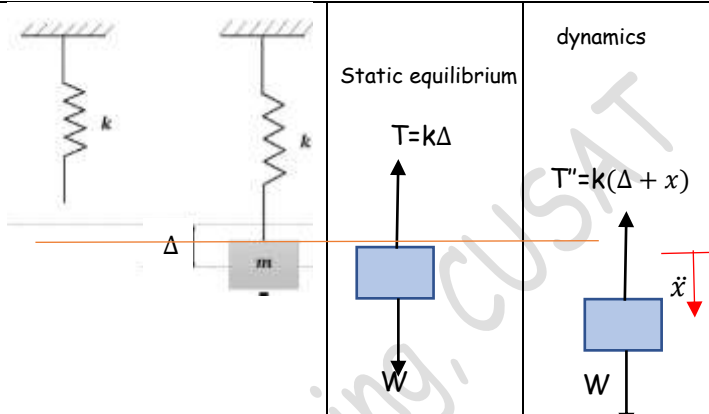


Force proportional to displacement

We have seen cases where the force is a constant, proportional to time, or proportional to velocity. Accordingly, we get acceleration as a constant, function of time, or function of velocity. Now let us consider another case where the force is proportional to displacement. Naturally, in such cases the acceleration will be also proportional to displacement. Such problems are termed as free vibrations or simple harmonic motion.

Let us consider a system where a block of weight W is suspended in a spring of stiffness k . When the block is suspended at the end of the spring, it will deform by an amount Δ (called static deflection) and come to an equilibrium position. If we draw the free-body diagram of the block in equilibrium, we can write the equation of static equilibrium as

$$W - k\Delta = 0$$



Now imagine that the block is pulled down and released, the system will move up and down. Consider an instant when the block moves x distance down from the equilibrium position. Assume that the block has an acceleration $\ddot{x}$ down ward

The equation of motion is

$$\frac{W}{g} \ddot{x} = W - k(\Delta + x) = W - k\Delta - kx = -kx$$

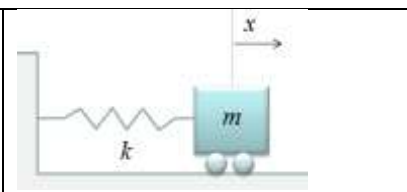
$$\ddot{x} = -\frac{kg}{W} x = -p^2 x$$

Where $p^2 = \frac{kg}{W}$ or $p = \sqrt{\frac{kg}{W}} = \sqrt{\frac{k}{m}}$. The acceleration is proportional to displacement.

$$\ddot{x} + p^2 x = 0$$

This is a second order, linear, homogeneous, ordinary differential equation with constant coefficients.

Consider the spring mass system in the horizontal direction. It is left as an exercise to derive the differential equation of motion. You will get the same equation of motion



The solution of this equation is

$$x(t) = A \sin pt + B \cos pt$$

The velocity is given by $\dot{x}(t) = Ap \cos pt - Bp \sin pt$. The acceleration $\ddot{x}(t) = -Ap^2 \sin pt - Bp^2 \cos pt$. A and B are arbitrary constants which can be evaluated from the initial conditions.

Let us consider a set of initial conditions: when $t = 0$, $x = x_0$, and $\dot{x} = 0$ (This means that the block is pulled down an amount x_0 and released.)

$$x_0 = A \times 0 + B \times 1 \gggg B = x_0$$

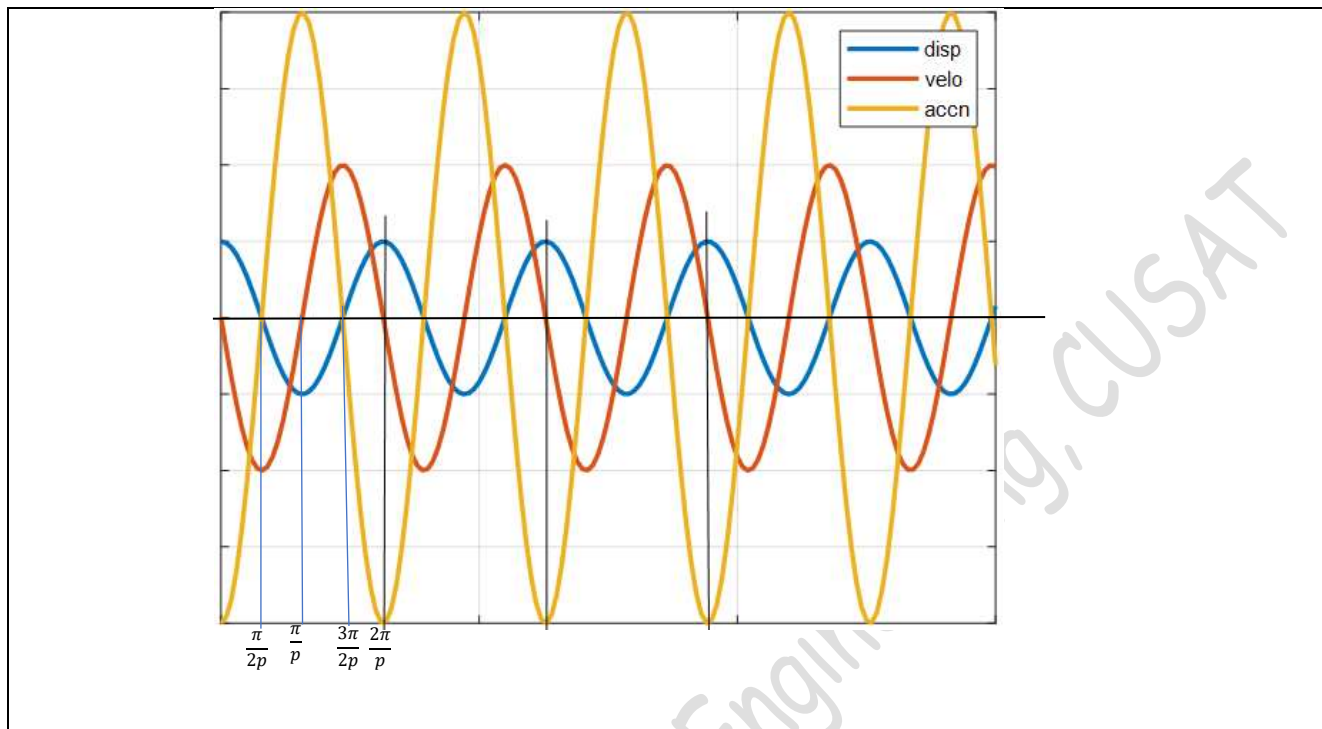
$$0 = Ap - B \times 0 \gg A = 0$$

$$\therefore x(t) = x_0 \cos pt$$

$$\dot{x}(t) = -x_0 p \sin pt$$

$$\ddot{x}(t) = -x_0 p^2 \cos pt$$

Let us plot the displacement, velocity, and acceleration against time. When $t = \frac{2\pi}{p}$, the plots complete one cycle. Hence the period of oscillation of the system $T = \frac{2\pi}{p} = 2\pi \sqrt{\frac{W}{kg}}$



It is left as an exercise to evaluate the arbitrary constants A and B and plot the displacement, velocity, and acceleration when the initial conditions are

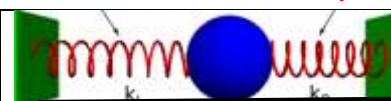
when $t = 0, x = 0$, and $\dot{x} = \dot{x}_0$

when $t = 0, x = x_0$, and $\dot{x} = \dot{x}_0$

In all the cases, we can see that the period of oscillation is $T = \frac{2\pi}{p}$. Hence, in the problems where the force is proportional to displacement, the system will have oscillations. In such cases, we write the equation in the form $\ddot{x} + p^2x = 0$. Then, the period of oscillations will be $T = \frac{2\pi}{p}$.

Example 1.

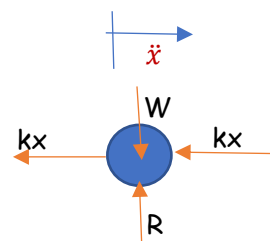
If two springs each having a stiffness k is attached to a ball of weight W. If the ball is initially displaced 1 cm to the right and released, find the period of oscillation of the ball. What is the maximum velocity? Assume that there is no friction between the ball and the floor.



Let the ball be displaced 1 cm to the right and released. The mass will oscillate. Let us consider an instant when the ball is x distance to the right and draw the free body diagram.

The forces are

1. The spring forces kx on right and left
2. Weight W
3. Normal reaction R



In the vertical direction, $\Sigma F_y = 0 \gggg R - W = 0 \gggg R = W$

In the horizontal direction, $\frac{W}{g} \ddot{x} = \Sigma F_x, \quad \frac{W}{g} \ddot{x} = -kx - kx = -2kx$

$$\ddot{x} + \frac{2kg}{W} x = 0$$

This is now of the form $\ddot{x} + p^2 x = 0$, where $p^2 = \frac{2kg}{W}$. Then, the period of oscillation

$$T = \frac{2\pi}{p} = \frac{2\pi}{\sqrt{\frac{2kg}{W}}} = 2\pi \sqrt{\frac{W}{2kg}}$$

The solution of the governing equation is $x(t) = A \sin pt + B \cos pt$

$$\dot{x}(t) = Ap \cos pt - Bp \sin pt$$

The initial conditions are when $t = 0$, $x = 1 \text{ cm}$, $\dot{x} = 0$

$$0.01 \text{ m} = A \times 0 + B \gggggggg B = 10^{-2}$$

$$0 = A \times p - B \times 0 \gggg A = 0$$

$$\dot{x}(t) = -\sqrt{\frac{W}{2kg}} \times 10^{-2} \times \sin pt$$

The maximum velocity $\dot{x}_{\max} = -\sqrt{\frac{W}{2kg}} \times 10^{-2} \frac{\text{m}}{\text{s}}$

Example 2

A small pan suspended from a helical spring as shown has an observed period of vibration T_0 . When carrying a known weight P , the observed period of vibration T_1 and when carrying an unknown weight W (without P), the observed period of vibration is T_2 . Find the magnitude of the weight W .



Let us assume that the weight of the pan be Q . When the pan is empty, the period of oscillation is given as T_0 .

$$T_0 = 2\pi \sqrt{\frac{Q}{kg}} \dots\dots\dots (1)$$

When a known weight P is added, the period is T_1

$$T_1 = 2\pi \sqrt{\frac{Q + P}{kg}} \dots\dots\dots (2)$$

When an unknown weight W is added, the period is observed as T_2

$$T_2 = 2\pi \sqrt{\frac{Q + W}{kg}} \dots\dots\dots (3)$$

$$T_1^2 - T_0^2 = 4\pi^2 \frac{P}{kg}$$

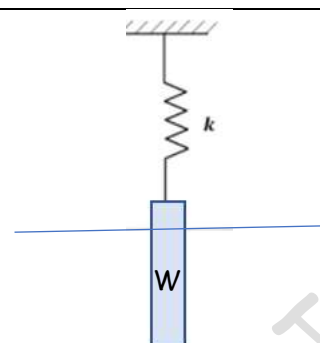
$$T_2^2 - T_0^2 = 4\pi^2 \frac{W}{kg}$$

$$\frac{T_2^2 - T_0^2}{T_1^2 - T_0^2} = \frac{4\pi^2 \frac{W}{kg}}{4\pi^2 \frac{P}{kg}} = \frac{W}{P}$$

$$W = \left(\frac{T_2^2 - T_0^2}{T_1^2 - T_0^2} \right) P$$

Example 3

A solid right circular cylinder of weight $W=45$ N and cross-sectional area $A=1.5 \times 10^{-2} \text{ m}^2$ is suspended by a spring of constant $k=175$ N/m and hangs partially submerged in water of specific weight $w=9800$ N/m³. Neglecting inertia of water, find the period of small oscillations.



Since the cylinder is hung vertically with the help of spring. There will be a static deflection of the spring. Let the length of cylinder submerged in water be h . Consider the static equilibrium of the cylinder.

The forces are

1. The weight of the cylinder, W .
2. The spring force, $k\Delta$
3. The upward buoyant force, B = weight of water displaced

$$B = Ahw$$

The equation of static equilibrium

$$Ahw - W + k\Delta = 0$$

static equilibrium

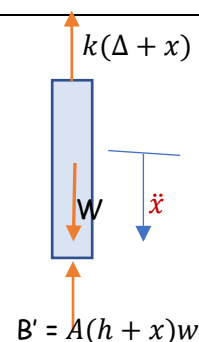


Let the cylinder be pulled down and released. The cylinder will oscillate up and down. Consider an instant when the cylinder moves down x distance from the equilibrium position. The acceleration is considered downward. The spring elongates $(\Delta + x)$, and the length of cylinder in water is $(h + x)$

Let us write the equation of motion in the vertical direction

$$\frac{W}{g} \ddot{x} = W - A(h + x)w - k(\Delta + x) = (W - Ahw - k\Delta) - (Aw + k)x$$

$$\frac{W}{g} \ddot{x} = -(Aw + k)x$$



$$\ddot{x} + \frac{g(Aw + k)}{W} x = 0$$

This is of the form $\ddot{x} + p^2 x = 0$, where $p^2 = \frac{g(Aw + k)}{W}$. Then, the period of oscillation

$$T = \frac{2\pi}{p} = \frac{2\pi}{\sqrt{\frac{g(Aw + k)}{W}}} = 2\pi \sqrt{\frac{W}{g(Aw + k)}} = 2\pi \sqrt{\frac{45}{9.81(1.5 \times 10^{-2} \times 9800 + 175)}} = 0.75 \text{ s}$$

Exercise 1

If 50 N tension produces an elongation of 25 mm in a given spring, find the frequency of vibration of a 5N weight suspended from the end of the spring.

Exercise 2

A 50 N weight suspended vertically by a spring vibrates with an amplitude of 75 mm and a frequency of 60 oscillations/minute. Find the spring constant, the maximum tension induced in the spring, and the maximum velocity of the weight.

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